Category

The Term Structure of Interest Rates: Spot, Par, and Forward Curves

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MATURITY STRUCTURE OF INTEREST RATES AND SPOT RATES

Learning Outcome

- define spot rates and the spot curve, and calculate the price of a bond using spot rates

Maturity Structure of Interest Rates

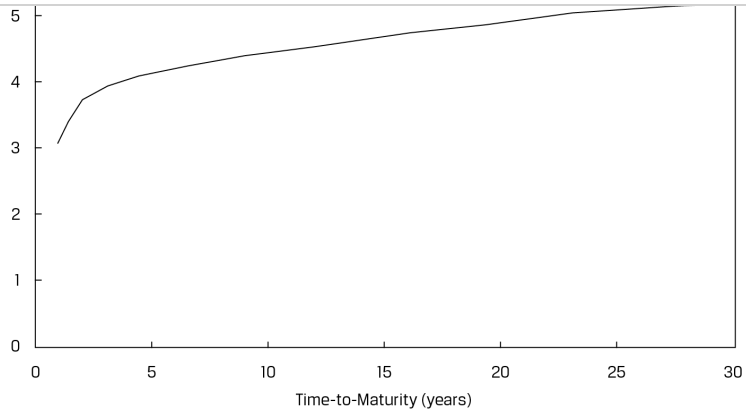
Suppose that the yield-to-maturity is higher on one bond compared to another bond. There are several possible reasons for the difference, including credit risk, different currencies, liquidity, tax differences, and the periodicity assumption used in the yield calculation. Credit risk and currency differences were demonstrated in earlier lessons where Romania's (Ministry of Finance) 30-year euro-denominated bonds were issued at a 4.657% yield-to-maturity, a 411 bp spread over Federal Republic of Germany bonds. Another reason bonds may have different yields-to-maturity is that they have different times-to-maturity. This factor explaining the differences in yields is called the **maturity structure of interest rates** or **term structure of interest rates**.

Term structure is best analyzed using bonds that have all the same properties *other than* time-to-maturity; that is, the bonds should be denominated in the same currency and have the same credit risk, liquidity, tax status, and periodicity assumption and they should have the same coupon rate so that they each have the same degree of coupon reinvestment risk. This ideal dataset would be yields-to-maturity on a series of default-risk-free *zero-coupon* bonds, known as **spot rates**, for a full range of maturities. Developed market sovereign bonds are typically used for this purpose, because they represent the lowest default risk among issuers in a given market. Collectively, this dataset is the government bond **spot curve**, sometimes called the zero or "strip" curve (because the coupon payments are "stripped" off the bonds).

The government bond spot curve is ideal for analyzing maturity structure because it best meets the "other things being equal" assumption. These government bonds presumably have the same currency, credit risk, liquidity, and tax status. Most importantly, as zero-coupon bonds, they have no coupon reinvestment risk. A government bond spot curve is illustrated in Exhibit 1 for maturities ranging from 1 to 30 years. The annual yields are stated on a semiannual bond basis, which facilitates comparison to coupon-bearing bonds that make semiannual payments, like corporate bonds in many markets.

Exhibit 1:

Government Bond Spot (Zero) Curve



The spot curve in Exhibit 1 is upward sloping and flattens for longer times-to-maturity, meaning that the longer-term government bonds have higher yields than the shorter-term bonds. This pattern is typical under normal market conditions. Sometimes, a spot curve is downward sloping in that shorter-term yields are higher than longer-term yields. Such downward-sloping spot curves are called inverted yield curves. Theories that attempt to explain the shape of the yield curve and its implications for future financial market conditions are covered later.

While a spot curve comprised of zero-coupon government bonds is ideal for analysis, there are several practical issues to contend with. First, most actively traded government bonds make coupon payments. These coupon bonds might not have the same liquidity and tax status. Older ("seasoned") bonds tend to be less liquid than newly issued debt because they are owned by "buy-and-hold" institutional and retail investors. Governments issue new debt for regular times-to-maturity—for instance, 5-year and 10-year bonds. The current 6-year bond could be a 10-year bond that was issued four years ago. Also, because interest rates fluctuate, older bonds can be priced at a discount or premium to par value, which can lead to tax differences because some tax jurisdictions distinguish capital gains and losses from interest income.

In practice, therefore, only the most recently issued and actively traded government bonds are used to build a yield curve. These bonds have similar liquidity, and because they are priced closer to par value, they have fewer tax effects. Additionally, because there are limited data for the full range of maturities, interpolation is used to complete the curve. Exhibit 2 illustrates a yield curve for a government that issues 2-year, 3-year, 5-year, 7-year, 10-year, and 30-year bonds that make semiannual coupon payments. Straight-line interpolation is used between those points on the yield curve.

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Exhibit 2:

Government Coupon Bond Yield Curve

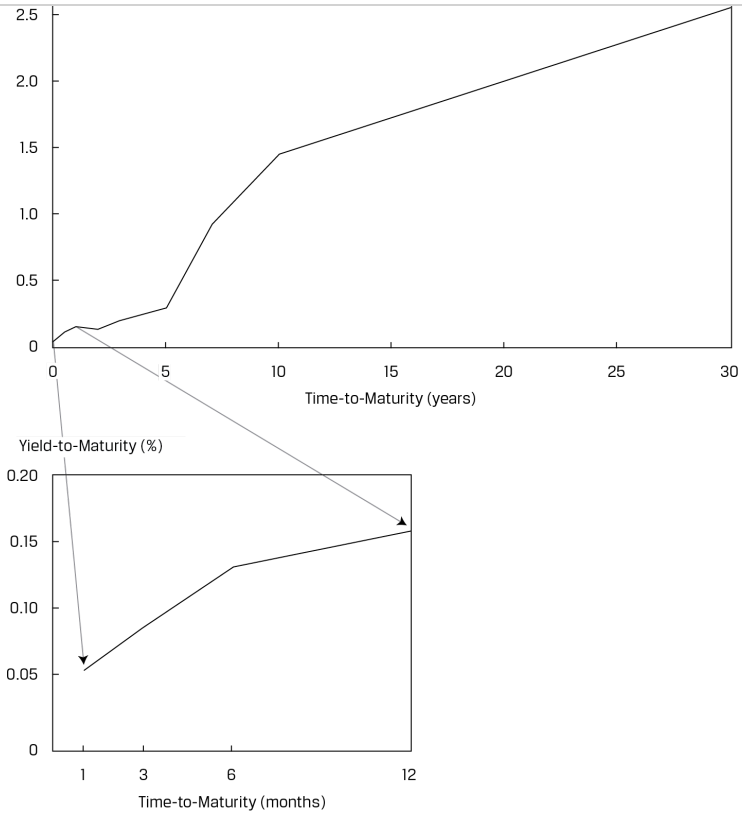


Exhibit 2 also includes yields for short-term government securities having 1 month, 3 months, 6 months, and 12 months to maturity. Although these money market instruments are issued and traded on a discount rate basis, they are converted to bond equivalent yields (i.e., add-on rates with a 365-day year). All bonds on the yield curve should also be converted to the same periodicity in order to facilitate comparisons across times-to-maturity. Conversion from discount rates to bond equivalent yields and across periodicities was demonstrated in prior lessons.

Bond Pricing Using Spot Rates

Since interest rates vary by maturity and have a term structure, we can price bonds using a sequence of discount rates that correspond to the cash flow dates. One such sequence, representing default-risk-free rates, is the spot curve. Bond prices determined using spot rates establish "no-arbitrage" prices; if a bond's price differs from its no-arbitrage value, an arbitrage opportunity exists in the absence of transaction costs.

Suppose that the one-year spot rate is 2%, the two-year spot rate is 3%, and the three-year spot rate is 4%. Then, the price of a three-year bond that makes a 5% annual coupon payment is 102.960:

$$\frac{5}{(1.02)^1} + \frac{5}{(1.03)^2} + \frac{105}{(1.04)^3} =$$

4.902 + 4.713 + 93.345 = 102.960.

This three-year bond is priced at a premium, so its yield-to-maturity must be less than its 5% coupon rate. Using the basic bond pricing formula, the yield-to-maturity is 3.935%:

$$102.960 = \frac{5}{(1+r)^1} + \frac{5}{(1+r)^2} + \frac{105}{(1+r)^3}; r = 0.03935.$$

If the cash flows are discounted using the yield-to-maturity, the same price is obtained:

$$\frac{5}{(1.03935)^1} + \frac{5}{(1.03935)^2} + \frac{105}{(1.03935)^3} =$$

4.811 + 4.629 + 93.520 = 102.960.

3.935%. The present value of the final cash flow, which includes the redemption of principal, is 93.345 at 4% and 93.520 at 3.935%. Nevertheless, the sum of the present values using either approach is 102.960.

Equation 1 is a general formula for calculating a bond price given the sequence of spot rates. Note that for pricing a bond with a different risk profile than the spot curve (such as credit risk, as for corporate bonds, for example), a spread would have to be added to the spot rates.

$$PV = \frac{PMT}{(1+Z_1)^1} + \frac{PMT}{(1+Z_2)^2} + \cdots + \frac{PMT+FV}{(1+Z_N)^N}, \qquad 1$$

where

Z_1 is the spot rate, or zero-coupon yield or zero rate, for period 1

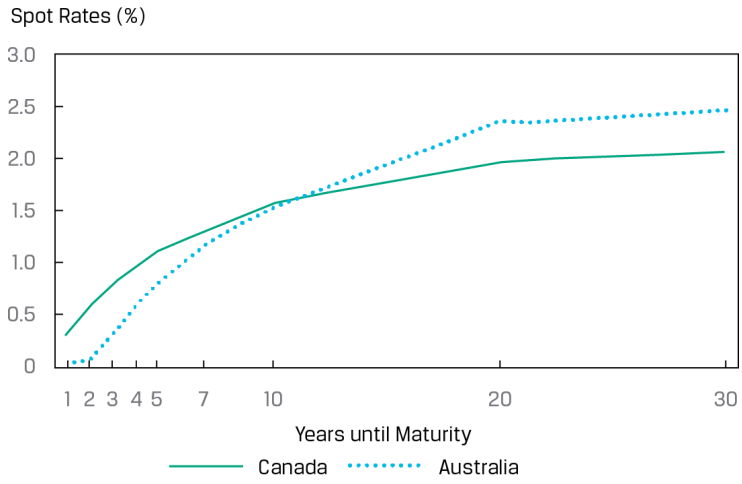
Z_2 is the spot rate, or zero-coupon yield or zero rate, for period 2

Z_N is the spot rate, or zero-coupon yield or zero rate, for period N

We'll use spot rates on Canadian and Australian government bonds to demonstrate bond pricing with spot rates.

Exhibit 3:

Government Bond Spot Rates: Canada and Australia



Maturity (Years)	1	2	3	4	5	7	10	20	30
Canada	0.31%	0.57%	0.80%	0.96%	1.11%	1.30%	1.58%	1.98%	2.06%
Australia	0.03%	0.07%	0.30%	0.59%	0.81%	1.17%	1.52%	2.35%	2.47%

EXAMPLE 1

Bond Prices and Yields-to-Maturity Based on Spot Rates

1. Calculate the price (per 100 par value) of a five-year Canadian government bond with a 1.00% coupon given the spot rates shown in Exhibit 3.

Hide Solution

Solution:

99.50. To calculate the price of the bond, we use Equation 1 and the spot rates provided for each cash flow.

$(1+0.0031)$ $(1+0.0057)$ $(1+0.0080)$ $(1+0.0096)$ $(1+0$

PV = 99.50.

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2. Calculate the price (per 100 par value) of a five-year Australian government bond with a 0.80% coupon given the spot rates shown in Exhibit 3.

Hide Solution

Solution:

100.01. To calculate the price of the bond, we use Equation 1 and the spot rates provided for each cash flow.

$$PV = \frac{PMT}{(1+Z_1)^1} + \frac{PMT}{(1+Z_2)^2} + \cdots + \frac{PMT+FV}{(1+Z_N)^N}.$$

$$PV = \frac{0.80}{(1+0.0003)^1} + \frac{0.80}{(1+0.0007)^2} + \frac{0.80}{(1+0.0030)^3} + \frac{0.80}{(1+0.0059)^4} + \frac{10}{(1+0}$$

PV = 100.01

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3. Calculate the YTM of the five-year Canadian government bond, using the price obtained in Question 1.

Hide Solution

Solution:

1.104%. Instead of solving for the price, we use Equation 1 to solve for the YTM, the IRR that equates the price computed in Question 1 with the present value of future cash flows. In Microsoft Excel or Google Sheets, the RATE or IRR function can be used to solve for *r*.

$$PV = \frac{PMT}{(1+r)^1} + \frac{PMT}{(1+r)^2} + \cdots + \frac{PMT+FV}{(1+r)^N}$$

$$99.50 = \frac{1}{(1+r)^1} + \frac{1}{(1+r)^2} + \frac{1}{(1+r)^3} + \frac{1}{(1+r)^4} + \frac{101}{(1+r)^5} =$$

r = 0.011036 = 1.104%

Notice that the yield of the coupon bond is slightly below the five-year spot rate.

Question Set

1. True or false: In practice, a spot curve is created by calculating the yields-to-maturity on recently issued coupon-paying government bonds of varying maturities.

A. True
B. False

Hide Solution

Solution:

True. While zero-coupon government risk-free bonds would be preferable, in practice, the spot curve is created using yields on recently issued coupon-paying government bonds and linear interpolation. These bonds best satisfy the conditions for term structure analysis.

Hide Solution

Solution:

Because there are usually limited data to calculate yields for the full range of maturities, it is usually necessary to interpolate between observed yields to complete curve data.

3. Calculate the price and yield-to-maturity of a 2.5% coupon, five-year bond given the following spot rates. Assume annual compounding.

1-year	3.000%
2-year	2.800%
3-year	2.600%
4-year	2.400%
5-year	2.200%

Hide Solution

Solution:

The price is 101.314.

$$PV = \frac{PMT}{(1+Z_1)^1} + \frac{PMT}{(1+Z_2)^2} + \cdots + \frac{PMT+FV}{(1+Z_N)^N}.$$

$$PV = \frac{2.50}{(1+0.03)^1} + \frac{2.50}{(1+0.028)^2} + \frac{2.50}{(1+0.026)^3} + \frac{2.50}{(1+0.024)^4} + \frac{102.50}{(1+0.022)^5}.$$

PV = 101.34

The YTM is 2.22%.

$$PV = \frac{PMT}{(1+r)^1} + \frac{PMT}{(1+r)^2} + \cdots + \frac{PMT+FV}{(1+r)^N}$$

$$101.34 = \frac{2.50}{(1+r)^1} + \frac{2.50}{(1+r)^2} + \frac{2.50}{(1+r)^3} + \frac{2.50}{(1+r)^4} + \frac{102.50}{(1+r)^5}$$

r = 0.022195 = 2.22%

4. Suppose we observed the following term structure for Canadian government bonds. Calculate the price of a 0.50% coupon, three-year Canadian government bond.

Maturity	Yield-to-maturity
1-Year	1.1000%
2-Year	0.9500%
3-Year	0.7951%
4-Year	0.5700%
5-Year	0.3100%

Hide Solution

Solution:

The price is 99.126.

$$PV = \frac{PMT}{(1+Z_1)^1} + \frac{PMT}{(1+Z_2)^2} + \cdots + \frac{PMT+FV}{(1+Z_N)^N}.$$

$$PV = \frac{0.50}{(1+0.011)^1} + \frac{0.50}{(1+0.095)^2} + \frac{100.50}{(1+0.007951)^3}$$

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Question 4 above. Contrast the Canadian spot curves in Example 1 and Question 4 and explain why these two different yield curves produced the same pricing result.

Hide Solution

Solution:

Example 1 has an upward-sloping term structure, while in Question 4, the term structure is downward sloping. An important similarity is that, despite the vast difference in starting points (i.e., the one-year spot rates) between the two examples, the three-year spot rates are similar. In Example 1, the three-year rate is 0.7977%, and the three-year rate in Question 4 is 0.7951%. As a result of the similarity of the spot rates in the upward- and downward-sloping term structure examples, it is possible to exhibit the same price (and YTM) in upward- and downward-sloping term structure environments.

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CR As per my calculation the PV in the Ex 1 Q2 should be 99.99

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